EXCEL SALES ANALYTICS MINI PROJECT

Project Documentation — Sessions S01–S10

1. Project Overview

The Excel Sales Analytics Mini Project uses Microsoft Excel to analyze sales performance across regions, products, months, salespeople, customer types, and sales channels.

Objective: To clean and organize sales data, analyze business performance, create an interactive dashboard, perform What-If Analysis, and identify useful business insights.

Business Problem: Sales information needs to be organized and summarized so that management can understand revenue performance, identify successful products and regions, and make better business decisions.

2. Workbook Structure

The 2 workbooks contains the following sheets:

1. Main Project Worksheets

- Customers
- Employees
- Products
- Cleaned Sales
- What-If Analysis
- Dashboard

2. Template Worksheets

- Start Here
- Formula Practice
- Pivot Analysis
- Business Insights
- PowerQuery Log
- Data Dictionary
- What – IF analysis

Tools used: Microsoft Excel, Power Query, Excel Tables, formulas, PivotTables, PivotCharts, slicers, Timeline, Scenario Manager, Goal Seek, and Sparklines.

3. Data Cleaning and Power Query

Power Query was used to combine monthly sales CSV files and merge the Product_Master, Customer_Master, and Employee_Master data using matching ID fields.

The final dataset was loaded into an Excel Table named **Sales_Clean**.

The main data-cleaning activities included:

- Removing duplicate records.
- Trimming extra spaces.

- Standardizing text formatting.
- Correcting an incorrect date.
- Handling missing customer and product information.
- Creating an Order_Band helper column based on revenue ranges.
- Preparing a Data_Dictionary sheet to explain the dataset fields.

The Refresh All test was marked **YES** in the PowerQuery_Log sheet.

These steps helped organize the sales data for analysis.

4. PivotTables and Sales Analysis

PivotTables were created to summarize sales data from different business perspectives.

The main analyses included:

- Revenue by region.
- Revenue by product and Top 5 products.
- Monthly revenue trends.
- Revenue by category.
- Salesperson performance.
- Revenue as a percentage of Grand Total.
- Monthly revenue compared with targets.
- Average revenue and quantity analysis.
- Customer and order counts.
- Order counts by revenue band.

The Sum, Count, and Average value settings were used where appropriate. Monthly dates were grouped to analyze sales trends over time.

These PivotTables helped identify high-performing regions, products, categories, salespeople, and customer groups.

5. Interactive Dashboard

An interactive Excel dashboard was created to present the main sales results on one screen.

5.1 KPI Cards

The dashboard contains six KPI cards:

1. Total Revenue
2. Total Orders
3. Total Units Sold
4. Average Order Value
5. Top Region
6. Top Product

The total revenue is approximately **₹2.27 crore (₹22,719,903.09)**.

TOTAL REVENUE	TOTAL ORDERS	TOTAL UNITS SOLD	AVERAGE ORDER VALUE	TOP REGION	TOP PRODUCT
₹ 2,27,19,903.09	600	2732	37866.51	East	Laptop

5.2 Charts

Seven charts were created:

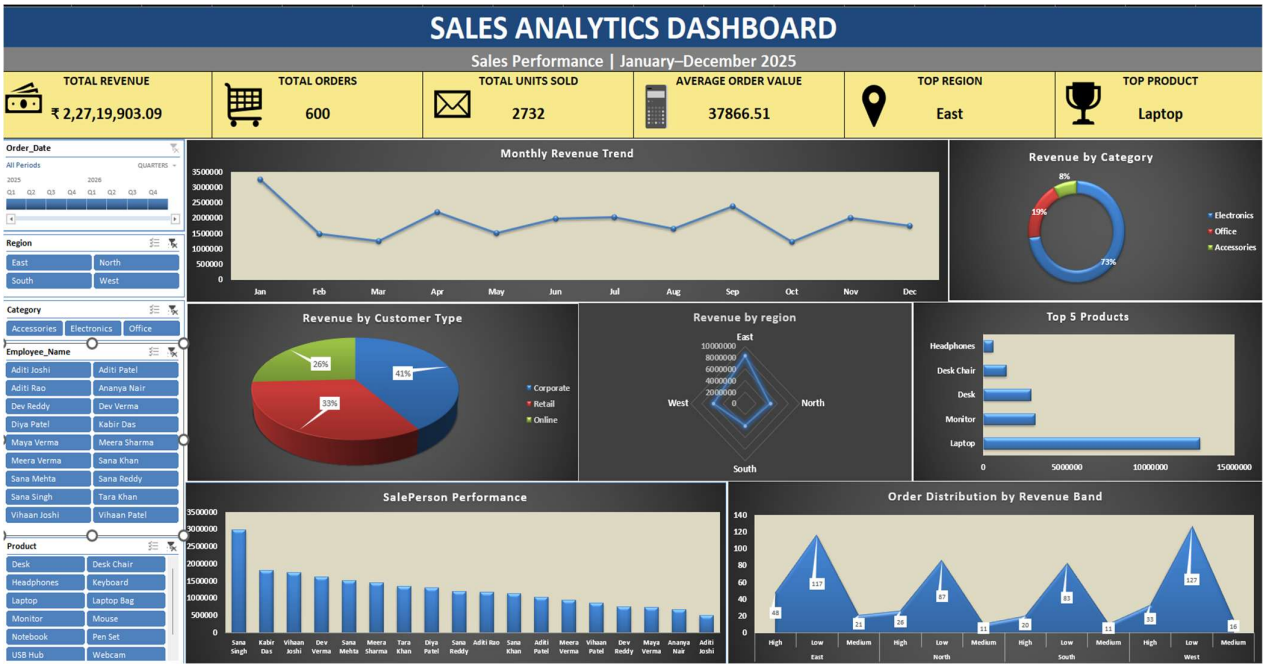
1. Revenue by Region
2. Monthly Revenue Trend
3. Revenue by Category
4. Top 5 Products
5. Salesperson Performance
6. Revenue by Category Type
7. Order Distribution by bands

5.3 Interactive Features

The dashboard includes:

- Region slicer.
- Category/Product slicer.
- Salesperson slicer.
- Product Slicer.
- Order Date Timeline.
- Report Connections for compatible PivotTables.

These features allow users to filter the dashboard and explore sales performance from different perspectives.



6. What-If Analysis

What-If Analysis was performed to understand how revenue changes under different assumptions. Scenario Manager was used to compare scenarios, and Goal Seek was used to determine the growth required to reach a target.

Scenario	Assumption	Projected Revenue
Base Case	0%	₹22,719,903.09
Conservative	-5%	₹21,583,907.94
Growth	+10%	₹24,991,893.40
Aggressive	+20%	₹27,263,883.71

Scenario Summary					
	Current Values:	Base Case	Conservative	Growth	Aggressive
Changing Cells:					
K11	-0.05	0	-0.05	0.1	0.2
Result Cells:					
K12	21583907.94	22719903.09	21583907.94	24991893.4	27263883.71

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

Goal Seek

The target revenue was **₹30,000,000 (₹3 crore)**.

Goal Seek calculated that approximately **32.04% growth** would be required to reach the target.

GOAL SEEK	
Base value	22719903.09
Assumption	32.04%
Projected Revenue	30000000

A monthly revenue Sparkline was also created to display the monthly revenue pattern in a compact visual format.

Month	Monthly Revenue	Monthly Revenue Trend
Jan	3255626.12	
Feb	1481863.5	
Mar	1250357.35	
Apr	2205023.13	
May	1500985.8	
Jun	1976621.35	
Jul	2022765.3	
Aug	1649154.97	
Sep	2371902.02	
Oct	1238809.8	
Nov	2008912.25	
Dec	1757881.5	

This analysis helps compare possible outcomes and supports revenue planning.

7. Business Findings and Recommended Actions

Finding 1: Highest-Revenue Region

Evidence: East has the highest revenue when no filters are applied.

Recommended action: Maintain the region's strengths and investigate growth opportunities in other regions.

Finding 2: Strongest Product

Evidence: Laptop is the top product in the Top 5 Products PivotTable.

Recommended action: Monitor stock levels and continue promoting this product.

Finding 3: Monthly Revenue Performance

Evidence: January has the highest revenue, while October has the lowest.

Recommended action: Review the factors that contributed to January's performance and investigate the decline in October.

Finding 4: Highest Average Revenue Category

Evidence: Electronics has the highest average revenue at approximately ₹78,772.02.

Recommended action: Maintain sufficient stock of Electronics products and explore ways to improve sales in other categories.

Finding 5: Salesperson Performance

Evidence: Sana Singh has the highest sales revenue.

Recommended action: Share effective sales practices with the team.

Finding 6: Customer Type Performance

Evidence: Corporate customers contribute the highest revenue.

Recommended action: Strengthen relationships with corporate customers and encourage repeat purchases.

Finding 7: Sales Channel Performance

Evidence: Phone is the best-performing sales channel.

Recommended action: Maintain good phone-sales support and customer service.

Finding 8: Contribution of Top Five Products

Evidence: The top five products generate ₹20,848,399.25, approximately 91.76% of total revenue.

Recommended action: Maintain the availability of these products while promoting other products to diversify revenue.

Finding 9: Revenue Scenarios

Evidence: Projected revenue is ₹21,583,907.94 under the -5% scenario, ₹24,991,893.40 under the +10% scenario, and ₹27,263,883.71 under the +20% scenario.

Recommended action: Use these scenarios to plan conservative, expected-growth, and aggressive revenue targets.

Finding 10: Data Quality

Evidence: Data preparation included removing duplicates, trimming spaces, standardizing text, correcting an incorrect date, handling missing customer and product information.

Recommended action: Apply data validation and conduct regular data-quality checks to improve accuracy.

8. One-Page Business Summary

The Excel Sales Analytics Mini Project was developed to organize sales information and analyze business performance using Microsoft Excel.

Power Query was used to combine monthly sales files and enrich the transaction data with product, customer, and employee information. Data cleaning addressed duplicates, extra spaces, text inconsistencies, an incorrect date, missing customer and product information, and a product-category correction.

PivotTables and PivotCharts were used to analyze revenue by region, product, category, month, salesperson, and customer type. An interactive dashboard was developed with six KPI cards, seven charts, slicers, and a date Timeline.

The total revenue was approximately ₹2.27 crore. East was the leading region, Laptop was the top product, January had the highest monthly revenue, and October had the lowest. Electronics had the highest average revenue at approximately ₹78,772.02. Corporate customers contributed the highest revenue by customer type, and Phone was the best-performing sales channel.

The top five products generated approximately 91.76% of total revenue. What-If Analysis showed the possible impact of revenue changes under -5%, +10%, and +20% scenarios. Goal Seek indicated that approximately 32.04% growth would be required to reach ₹3 crore.

The findings suggest maintaining leading-product availability, strengthening corporate customer relationships, improving weaker sales periods and regions, and diversifying revenue across more products. Regular data-quality checks are also recommended.

Overall, the project demonstrates how Excel can transform sales data into meaningful summaries and practical business recommendations.

9. Session Status

- **S01–S02:** Completed as instructor-assigned formula practice tasks, separate from the main project.
- **S03–S05:** Continued learning and practice tasks; not included as completed main-project deliverables.
- **S06–S07:** PivotTables, PivotCharts, dashboard, slicers, Timeline, and Report Connections completed.
- **S08:** Power Query preparation completed; Refresh All test marked YES.
- **S09:** Dashboard KPIs, Sparkline, Scenario Manager, and Goal Seek completed.
- **S10:** Business Insights and project analysis completed; the business summary is included in this documentation.

10. Conclusion

The Excel Sales Analytics Mini Project demonstrates how data cleaning, Power Query, PivotTables, PivotCharts, dashboard interactivity, and What-If Analysis can be used to understand sales performance.

The analysis identified leading regions, products, customer groups, salespeople, and channels. Scenario analysis also helped evaluate possible revenue outcomes and estimate the growth required to achieve the target.

Regular data-quality checks and review of the known Top Product KPI limitation will help improve the reliability of the workbook. Overall, the project provides a practical example of using Excel for sales analysis and business decision-making.